DATA INNOVATION FOR AFRICA (DI4A) INITIATIVE

Establishing a Data Innovation Lab

Applying the AfDB's Data Innovation Maturity Model

Based on the practical guide by the AfDB Statistics Department (October 2025)

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01

Context

Why must official statistics innovate?



Five pressures are converging on African NSOs



Data deluge

Official statistics must defend their visibility against faster but less reliable data.



Timeliness gap

Users want immediate, granular data; rigour takes time.



Fast-moving technology

Tools evolve constantly, requiring monitoring, agility and resources.



Scarce skills

Few data scientists in the public sector, facing private-sector competition.



Tight budgets

Strained national budgets; short-term, narrowly targeted aid.

Innovate or die: transformation is now a condition for relevance.

Modernising is no longer enough: transform

System modernisation

Purpose	Quality and accessibility of official statistics
Nature	Internal, institutional
Who	NSS members
Key issues	Reliability, independence

Ecosystem transformation

Purpose	An open, collaborative environment around data
Nature	Global, systemic
Who	All data holders and users
Key issues	Interoperability, governance, innovation

Reach beyond the NSS

Cross-cutting innovation

Co-create solutions

Innovation at every stage of the value chain



Sources

- Satellite imagery
- Mobile phone data
- Administrative records

Collection

- CAPI
- Web scraping
- Internet of things

Processing

- Automated validation
- Machine learning
- Cloud pipelines

Storage

- Data catalogues
- Large-scale databases
- Cybersecurity

Dissemination

- Open data portals
- APIs
- Dashboards

Analysis

- Predictive analytics
- Real-time monitoring
- AI, Python, R

Key principle: innovation starts with the technicians who know the tools and processes.

02

The AfDB DIL initiative

What does the Bank offer to accelerate innovation?



Four objectives, two pillars, four principles

OBJECTIVES

1 Enable innovation

Within the Bank and Regional Member Countries

2 Augmented data

Faster, more granular and cheaper

3 Demystify data science

Through training and joint projects

4 Open-source tools

A tested, reusable repository

PILLARS

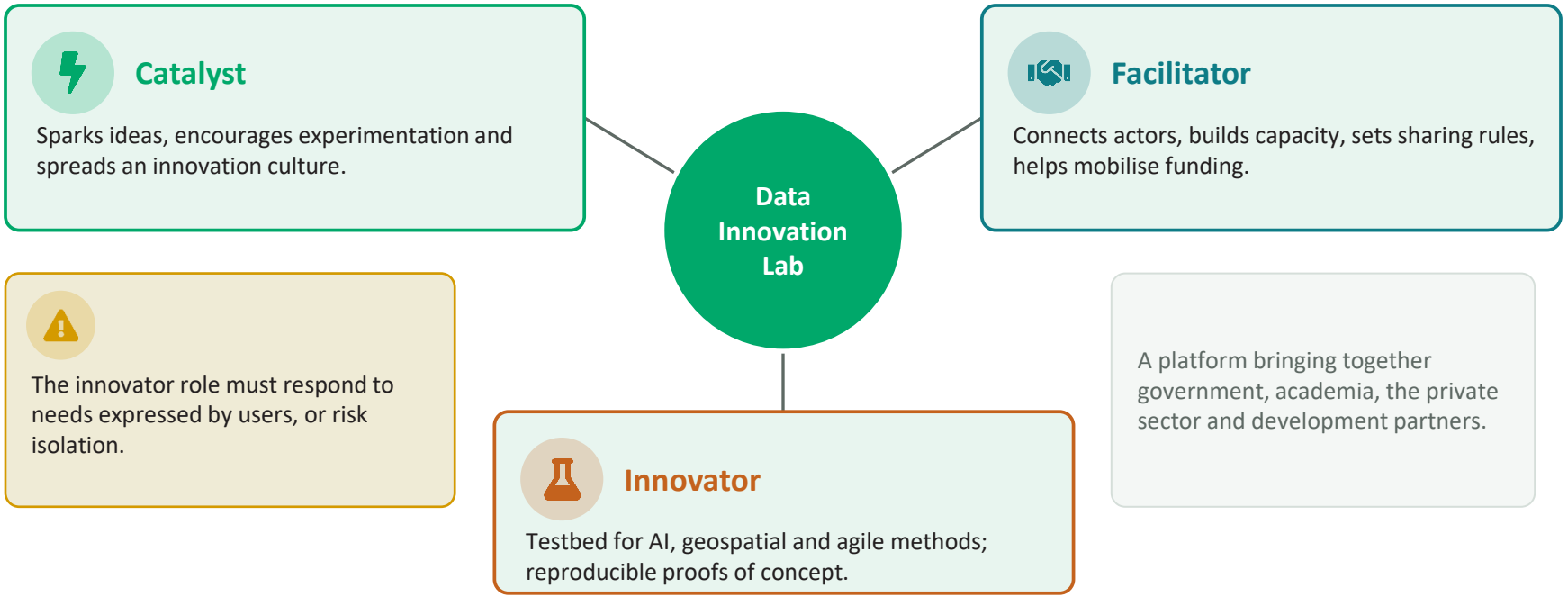
1 Innovation for value and impact

2 Networks and collaboration

PRINCIPLES

- Data augmentation
- Collect once, reuse
- Reduce friction
- Partnership at the core

A DIL plays three complementary roles



03

The maturity model

How ready is a data ecosystem to transform?



Twelve dimensions for a full diagnostic



94 open questions

adaptable to each country's context

No weighting, no ranking

a dialogue tool, not a benchmark

Supported first

first diagnostic with the AfDB, then country-led follow-ups

A participatory diagnostic, reported as a traffic light

1. **Questionnaire**
   Initial answers from the NSO and its partners
2. **Expert mission**
   Five to six days at most, to mobilise teams
3. **Colour coding**
   One colour per question, looking for trends
4. **Feedback**
   Technical analysis and summary for decision-makers



Green
Favourable situation

Orange
Room for improvement, sustained attention

Red
Further reflection and priority measures

Two formats: 12 dimensions for decision-makers, about 60 components for detail.

2–3 months per country, end to end

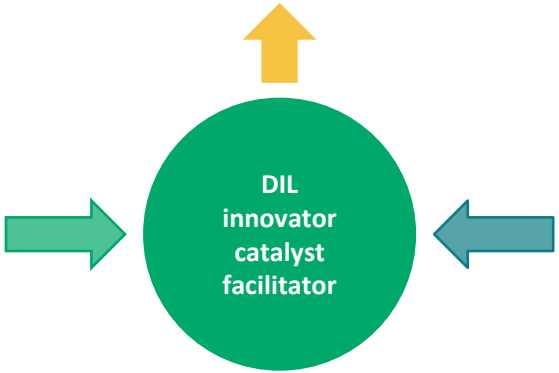
Two families of levers to transform the ecosystem

Supporting factors

Within the NSO's control

- Internal vision
- Human resources
- ICT infrastructure
- Innovative experiences
- Quality and standards

Transformation of the data ecosystem



The diagnostic is the first step

Steering factors

Less control, but decisive

- Legal framework
- Strategic framework
- Partnerships
- Communication and users
- Funding
- Data analytics

04

Implementation

What have the pilots taught us?



Physical or virtual DIL: a trade-off to make

Physical DIL

A dedicated unit, often under IT or methods

Advantages

- Visible in the organisation chart
- Clear staff assignments
- Simpler budget management

Risks

- The unit monopolises innovation
- Technical departments disengage

Virtual DIL

Shared responsibility across departments, light coordination

Advantages

- Whole institution engaged
- Investment close to operations

Risks

- Diluted responsibilities
- Loss of overall coherence

Two conditions: keep innovation close to operations and coordinate all investments.

Four lessons to remove bottlenecks



Introduce “innovation time”

Like training time, with a light committee reviewing projects and results.



Create a data-roles competency framework

Without civil-service status, data scientists remain hard to recruit.



Start with a core group of partners

Deliver tangible results with a few actors, then expand the coalition.



Start with existing infrastructure

Share public capacity, technology watch and even purchases.

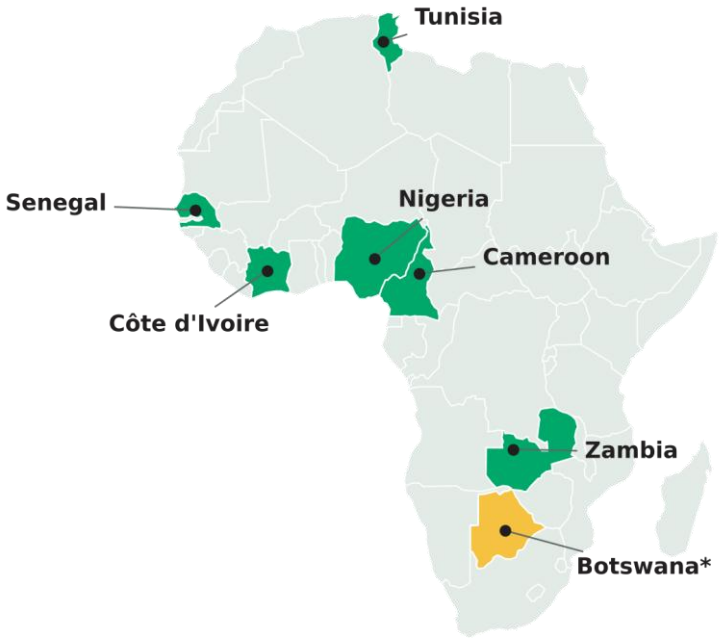
05

Country experiences

What the pilot countries are already doing



The model tested in six pilot countries



Nov 2023	Tunisia
Feb 2024	Senegal
Mar 2024	Côte d'Ivoire
Jun 2024	Nigeria
Sep 2024	Cameroon
Apr 2025	Zambia

** Botswana: listed only in Annex 7 of the guide.*

Map: Natural Earth (public domain). Dates: AfDB guide, October 2025.

Good practices across the value chain

Senegal

Use cases with ENSAE interns; mobile data to predict poverty indicators; algorithmic household targeting.

Tunisia

Mobile data for internal migration, validated against the census; partnership with the national remote-sensing centre.

Côte d'Ivoire

Talks with an operator on cost-sharing; Sentinel-2 then commercial imagery; an "innovation café".

Nigeria

A base of national strategies: data, data protection, cloud, interoperability, open data.

Cameroon

Workshop on satellite data for population statistics, open to the NSS, universities and private sector.

Web scraping

Consumer prices (Côte d'Ivoire, Cameroon, Senegal) and job vacancies (Côte d'Ivoire).

The AfDB DIL's activities in figures

127
participants · 22 countries
"Innovate Africa" symposium, Kigali 2024

48
participants · 13 countries
DIHAP health-data programme, 2024

351
communes analysed
Night-time lights (VIIRS), Burkina Faso

117
participants
"Brown Bag" webinar series, 2024

13
countries convened
AU STG17 group, Kigali, Sep 2025

2
agreements signed
Data licence agreements (Meta, JBA)

Source: AfDB guide, Annex 7 (activities 2022–2025).

06

Way forward

How can the initiative scale up?



Five directions for the initiative



01
Support national action plans

02
Expand country diagnostics

03
Support follow-up self-diagnostics

04
Strengthen country networks

05
Mobilise continental actors

Already under way: in September 2025 the AU STG17 group adopted a 2026–2030 action plan on big data and AI in official statistics.

Points of attention for a future edition

What the guide does not yet say

- No criteria for assigning colours
- No monitoring and evaluation indicators
- No cost estimates
- Ethics, privacy and generative AI barely covered
- Some inconsistent dates and figures

Suggested improvements

- A scoring rubric per question
- A results framework with indicators
- A typical three-year DIL budget
- A data-protection and AI component
- Editorial review and harmonisation

Sources

African Development Bank, Statistics Department (October 2025)

Establishing a Data Innovation Lab: A Practical Guide to Applying the AfDB's Data Innovation Maturity Model.

afdb.org/en/documents/data-innovation-africa-di4a-initiative-practical-guide-applying-afdbs-data-innovation-maturity-model

Natural Earth

1:110m cartographic data, public domain (pilot-country map).

naturalearthdata.com

Note on visual identity

Design inspired by the AfDB visual identity; this is not an official Bank document.

IN SUMMARY

Innovation starts by letting go of old ways of thinking

1. Diagnose with the twelve dimensions
2. Choose a fitting organisation
3. Build coalitions and measure progress

Thank you